queries (subsection 4.1) which extend the conjunctive (narrowing) queries prevalent in DSN tools.

Designers and Scientists. The evaluation clearly highlighted the different concerns that designers and building scientists had when using a DSN interface like DesignSense. In fact, Matejka et al. [MGB*18] made a similar distinction between the 'engineer' type of users and 'artists'. Matejka et al. also argue that accommodating both groups (and those in between) can encourage them to appreciate each other's concerns. The evaluation we have conducted contributes specific feature suggestions by each of the groups.

Navigation + Analysis. The evaluation revealed the need for better supporting the process of design decision-making and the demand for integrating design navigation with data analysis (i.e. Design Analytics). This can be accomplished through visually-assisted support for techniques such as sensitivity analysis and value function elicitation, among others.

Form and Data Coupling: A Bird-View. Although we focused our design study on the architectural design domain, the same need for coupling form and data and the same challenge of choice overload exists in other domains where forms are culturally important such as product design, mechanical design, urban planning and game design (subsection 7.4) to name a few. The demand for coupling *'form'* and data is not exclusive to design domains either, the design study by McCurdy and Meyer [MM19] with astronomers and astrophysicists showed a lack of support for the coordinated statistical and visual analysis of galaxy observations.

9. Conclusions and Future Work

In this work, we presented DesignSense, a design space navigation (DSN) interface for navigating large design spaces through interactive data visualization and analysis. We informed our design through collaboration with domain experts and a critical analysis of the current DSN solutions. We presented a formative focus group evaluation that suggested several areas for improvement, and two case studies to illustrate the utility of our interface.

The focus group lasted for only two hours and was mostly feature-based. A future evaluation should directly assess DesignSense with respect to the stated domain tasks. We also intend to release the tool for public use as well as hold close collaborations with our industry partners for using DesignSense on their Generative Design projects.

10. Case Study 1: Generative Design for Architectural Rezoning

One of the reasons for using generative design is widening exploration at the early design stages. Due to the open-ended nature of the generative process, the produced designs can be surprising to designers [CSL20]. Generative processes produce a large number of alternatives, which if properly generated and explored, can lead to considering far more alternatives within the same duration needed to consider a few alternatives in a non-generative design process.

An architectural design firm collaborating in this research project gave us access to a dataset resulting from a generative design process that they used in a past project. The dataset tackles an architectural design problem, namely a massing study, i.e., finding the masses or forms of a group of buildings that will be placed on a single site such that these buildings jointly meet certain requirements and maximize some objectives. The goal of this dataset is to explore different form compositions and spatial layouts of these forms in the building site. This dataset includes about 100 generated alternatives each representing a different massing solution and evaluated against multiple performance objectives.

A Sample. Figure 8 shows a sample from the dataset. The massing consists of the coloured buildings. The colours on the surfaces represent the solar radiation that the surfaces received at a single moment of time (red means more radiation). The gray buildings are part of the site (i.e., already built). This alternative includes three distinct masses. The space between the two on the right is secluded from sunshine, which is not necessarily good in the cold weather of Vancouver where the project is taking place. The tall block in the mass at the bottom blocks the views of the other masses to the south and south-east. This, in turn, blocks the sought winter sun that comes from the south. The metrics alone may not be informative except when compared against those of the other alternatives.

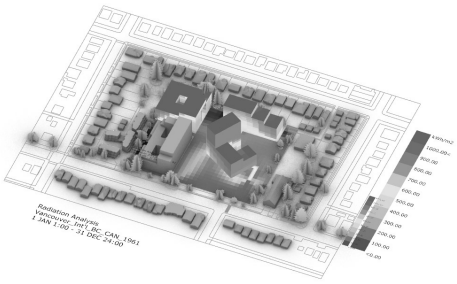


Figure 8: *A massing alternative.*

Quantitative Criteria. The objectives involved in this study can be found in the list below. Some of those (like solar radiation) are dependent on the geographic location of the project (Vancouver, Canada) and some are based on the form composition and its relation to the site. A number of these metrics require some form of aggregation across points on the site or on the building's surfaces. In these cases, the site/surfaces are discretized first and the average of calculating the metric for each discrete unit is reported.

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- FSR: after dividing each building mass into floors, FSR is the ratio between the area of these floors and that of the site.
- A/V ratio: the ratio between the surface area of all the building masses to their volumes.
- Site_Cvr%: the percentage of the site that has buildings on it.
- Roof_Solar, Horizontal_Solar: the annual solar energy gain on building's roofs and surfaces , respectively.
- SVF: the Sky View Factor is the ratio of the sky hemisphere visible from the ground.
- Isovist: the volume of space visible from a given point. In this case the point is taken to be a point on the ground.
- Sunshine_hrs: the annual hours of exposure to sunlight .
- Max_H: the height of the tallest building in the massing.
- Surface_View: the percentage of access to views outside the site to a chosen radius.

In general, this is an example of a sustainable design project where multiple interdisciplinary objectives are sought. Considering these objectives at an early stage often yields the highest returns, since some major (or macro) decisions can be made at this stage that can impact the eventual environmental footprint of the buildings. Furthermore, changing design alternatives is relatively cheap at this stage.

Qualitative Criteria. The following are some qualitative aspects that can be assessed by examining alternatives in either 2D images or a 3D geometry view: The symmetry of the massing, the availability of setbacks, accessibility (e.g., ease of traversal between buildings), empty spaces (how much they are and whether they can be well utilized), how "broken" the massing is (fewer and more connected building blocks vs. more in number but disconnected blocks), and the general complexity of the massing and consequently the ease of its construction.

Form and Data. Coupling the qualitative and quantitative aspects of alternatives can be achieved through coordinated selection or inspection.

Coordinated Inspection. To illustrate how inspection works we can look at the alternative we have described earlier. When inspecting an alternative, DesignSense highlights it in all the supported views. First, we see a large image view at the top of the gallery view. This view can also be switched to show a 3D model of the alternative or different 2D images of the same alternative (using the icons at the top-left of the large image). Below that is a gallery of sorted thumbnails, and the thumbnail of the inspected alternative is highlighted with thick borders. Note that the thumbnails here show only the selected alternatives, the crossed eye icon at the top bar indicates that. DesignSense allows inspection by hovering over the thumbnails in the gallery, lines in the PCP or the dots in the scatterplot.

Since the alternatives are sorted by their Site Coverage, we learn that this alternative achieved this target well since sorting is in descending order (notice the downward arrow the top bar) and the inspected alternative appears at the top of the gallery. When looking at the parallel coordinates plot, we find the exact metrics' values shown on the tooltips. We can compare this alternative's performance to others by observing the placement of the line along any of the vertical axes. For example, the Site Coverage has an average value when compared to all the alternatives in the dataset but is high when compared against the selected alternatives only.

Coordinated inspection can be used to ensure that design forms and data are matching. For example, we can ensure whether the site coverage metric is reliable by inspecting a few alternatives visually. We can do that by starting from the alternative with the lowest site coverage to the highest while keeping an eye on the large image in the gallery. If the metric is reliable, then we should observe that the amount of empty space on the site decreases as the coverage metric increases. Finally, we can achieve this by inspecting alternatives along the site coverage axis on the PCP starting from the bottom.

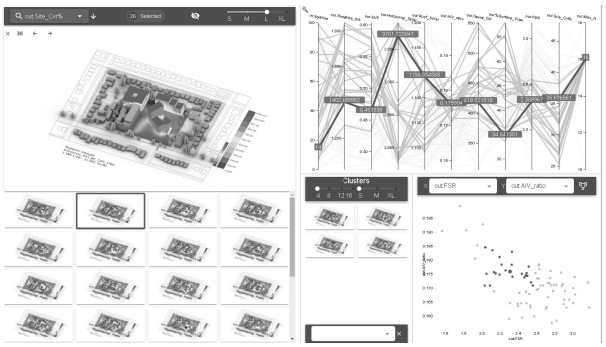


Figure 9: *A massing alternative under inspection.*

Coordinated Selection. When navigating design alternatives, we may have well-defined criteria for what we are looking for, or we form these criteria as we compare and examine alternatives. These criteria can be quantitative, qualitative or both. When a set of alternatives are selected in DesignSense, then the selection is coordinated across the interface. This allows us to select alternatives based on their quantitative performance (e.g., on the scatterplot view) and follow that with refining the selection based on the visual appearance (qualitative) of the alternatives in the selection. Alternating between quantitative and qualitative criteria is accomplished by alternating between selecting on data and form views.

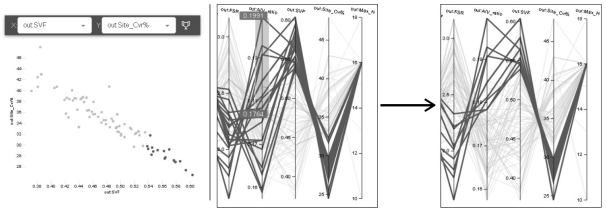


Figure 10: *The scatterplot on the left shows a strong negative correlation between SVF and Site Coverage. This is also clear from the crossing lines between the axes of these metrics on the PCP. The left PCP shows alternatives with high SVF already selected, and a tooltip shows the range on A/V ratio that will be selected once the user completes this operation. Since the current selection operation is Intersection, the result on PCP on the right shows alternatives meeting both criteria.*

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An example of clear criteria might be to find a set of massing alternatives that have a high SVF, a medium A/V ratio, and a 'simple' massing. Since the quantitative criteria are clear, we can start by applying it. First, we select the alternatives with a high SVF by brushing on the PCP. We do not want to quickly narrow the design space, as that will prevent us from flexing around trade-offs if they existed. For example, the site coverage metric negatively correlates with SVF (as indicated by the crossing lines in the PCP). If we selected only the alternatives with the highest SVF then we are limited to alternatives with very low site coverage. We might decide that a slightly lower SVF is appropriate if we found other reasons to motivate that. Either way, we do not have to strongly impose our criterion at the start so as to allow us to see and then consider these trade-offs. The SVF metric spans between 0.37 and 0.60. To follow this guideline, we can select the alternatives with SVF higher than 0.53. We can precisely select this range on the PCP using the tooltips that appear while brushing.

The next criterion concerns the A/V ratio. Inspecting a few alternatives along this metric's axis on the PCP shows that the masses tend to be simpler with higher values for this metric (and vice versa). We can find the intersection between the currently selected alternatives (with high SVF) and those with middle to high A/V ratio (Figure 10).

Upon filtering by data metrics, we can examine the design forms. First, we can enable the "Show only Selection" mode on the gallery so that only the thumbnails of the selected alternatives be shown (Figure 11). We can then inspect these thumbnails one by one or use their thumbnails in the gallery (if they were telling us enough) to further refine the selection by discarding the alternatives with complex masses.

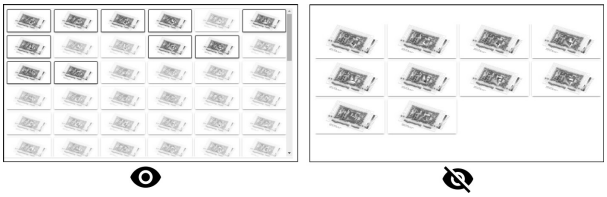


Figure 11: *Users can alternate between (left) showing the selected alternatives in the context of those unselected, or (right) show only the selected alternatives.*

Showing only the selected alternatives in the gallery is also effective at understanding the relation between forms and data (just like coordinated inspection, as described above). One can systematically select portions of the design space based on data queries and see only the forms in that part. The process can be repeated as needed to gain an understanding of the impact of data-based queries on the changes in the forms.

Finally, the example above shows filtering by data and form when clear criteria exist. When that is not the case, then we need to build our understanding of the design space first. We can do that by observing the correlations between the data metrics using the PCP or the scatterplot views. We can also scan through the design forms to shape an impression of the variety that exists. More what-if questions can be involved here than when we had clear criteria. This can be done by experimenting with selections. For example, an initial selection can be made on any of the data metrics. We can observe how the values of the other metrics are impacted by that as well as inspect the forms of the selected alternatives. Doing so allows us to understand the consequences of this selection. This process can be repeated, potentially by building more complex criteria using the intersection or union operations. Furthermore, our exploration might invoke any of the navigation strategies in DesignSense like zooming, clustering, or creating sets. The next sections will describe those in more detail.

Inspecting Cluster Representatives and Navigating with Clusters. Instead of examining the alternatives in the Augmented Gallery or the PCP views, users can choose to explore the dataset at a coarse level first. In other words, explore families of alternatives before examining individual alternatives. The cluster view can help in that. As a start, users may pick all or a few performance dimensions to cluster on. Then by inspecting the representatives, they gain an overview of the most distinct alternatives from a performance perspective. Inspection highlights an alternative on all the views (gallery, PCP, scatterplot), which allows users to compare representatives on all these views by quickly alternating between them (Figure 12). Lastly, picking a small number of clusters (e.g., K = 4) means that much of the variation is suppressed. Users can then systematically increase K such that representatives give a slightly finer and more nuanced view of the design space but not too fine that it defeats the purpose of clustering.

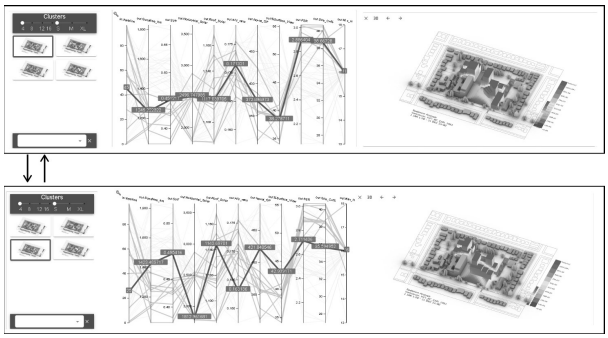


Figure 12: *By recurring inspecting cluster representatives, their performance and forms can be compared, and consequentially the clusters they represent.*

Another way to use the clusters view is to couple it with zooming (Figure 13). For example, upon clustering by the Area/Volume Ratio (A/V%) and the Site Coverage (SC%) into 4 cluster we get clusters of [high A/V%, low SC%], [low A/V%, high SC%], [mid A/V%, slightly high SC%], and [slightly high A/V%, mid SC%]. A user may decide that a low site coverage (first cluster) means under-utilizing the land available, which they do not wish to have. By clicking on each of the other clusters (while pressing the Addition selection modifier), they can collect the clusters they are more interested in and then zoom into the current selection. Alternatively, they can select the undesired cluster and exclude it.

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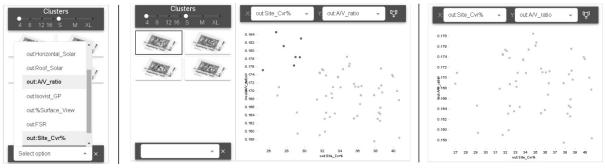


Figure 13: *Left: picking the dimensions to cluster on. Middle: selecting the cluster with a high A/V ratio and low site coverage (which is reinforced by looking at the scatterplot). Right: the dataset after excluding the cluster with low site coverage.*

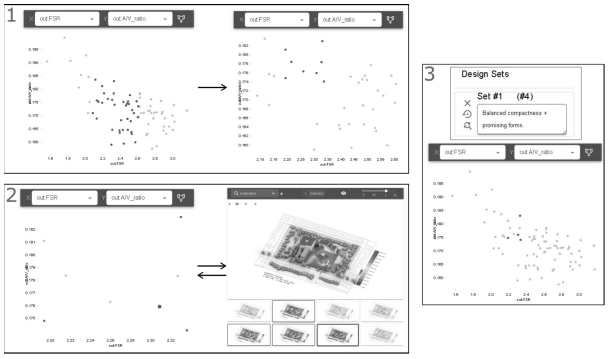


Figure 14: *(1) zooming down with data queries. (2) alternating between data and form selection. (3) resulting alternatives are saved as a set, then the zooming is reversed and the set is viewed in light of the larger dataset.*

Following a zoom or an exclusion operation, the clusters are recalculated again. Users can choose to repeat either of the above strategies (inspecting representatives or navigation with clusters) on the new clustering or switch to other means of exploration. A final note to make is that the number of clusters or the choice of dimensions to cluster on are visually guided by the exploration process instead of being performed as a pre-processing step.

Sets as Shopping Carts. After a line of inquiry leads to a smaller set of alternatives, users may want to save their progress to revisit it later or combine the results of this inquiry with earlier results. Sets can help with that. Saving progress is aided by the ability to associate comments with sets. Comments remind users of the content of the set or the criterion for creating it. Sets can also be combined using the selection operations of Intersection and Union. Combining sets allow us to connect criteria that may not be otherwise reconcilable. For example, a set may be created based solely on visual aesthetics and another based on the compactness of the buildings' forms (e.g., using the Area/Volume ratio). Finding the intersection of these sets amounts to finding alternatives meeting both criteria. If the intersection resulted in an empty set, then that may suggest that some conflicting objectives are at play or that the sets were simply not large enough.

Reflection with Sets. As users navigate, they make decisions about what to focus on and what to ignore. These decisions may lead to fixating on portions of the design space. By saving their progress as a set, users can then view it in light of all the other alternatives in the design space, leading them to recognize the trade-offs and decisions they committed to in the course of navigation. Figure 14 illustrates that.

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